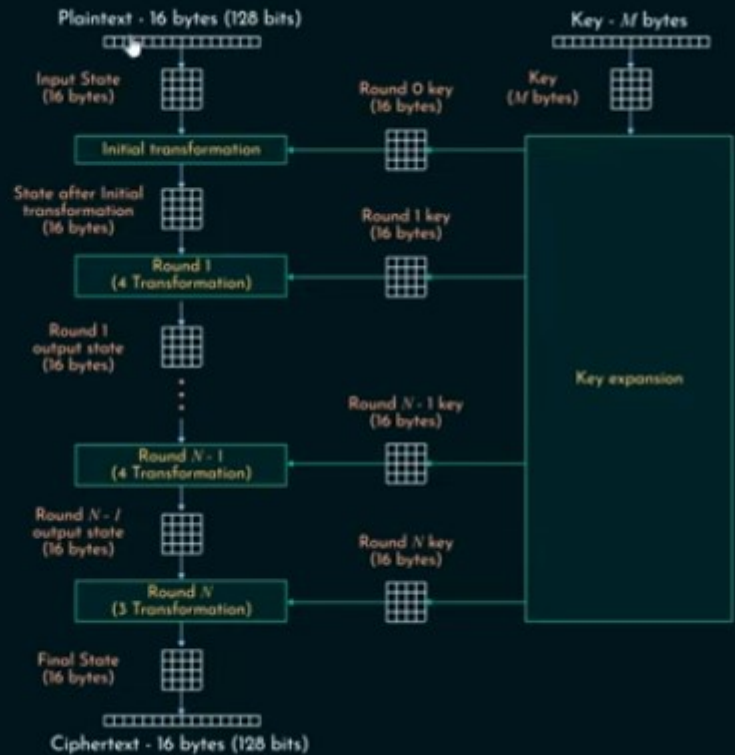


AES Structure



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AES Structure

No. of rounds	Key size (in bits)
10	128
12	192
14	256



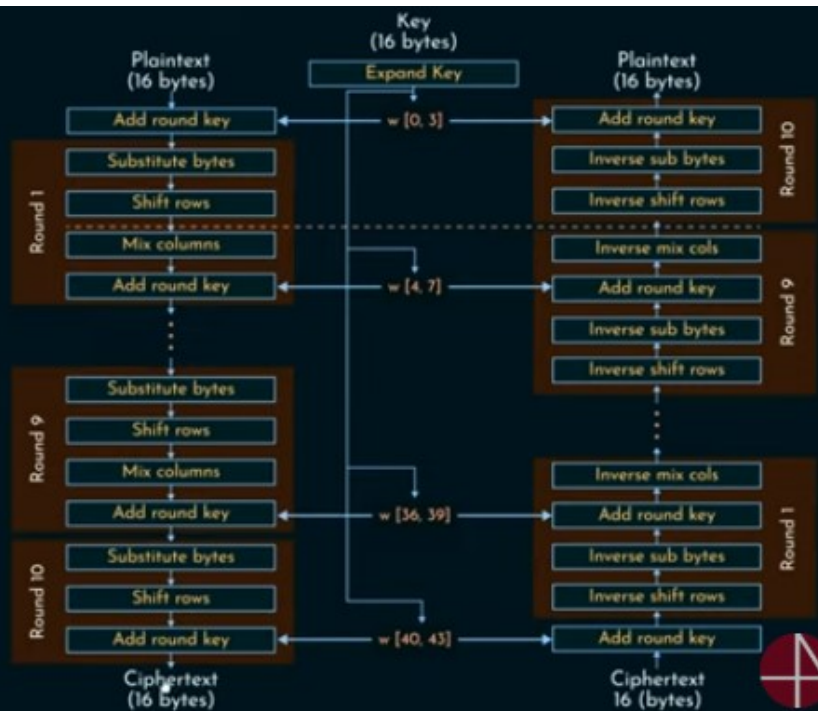
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AES Parameters

	AES-128	AES-192	AES-256
Key Size	128	192	256
Plaintext Size	128	128	128
Number of rounds	10	12	14
Round Key Size	128	128	128

AES Encryption and Decryption

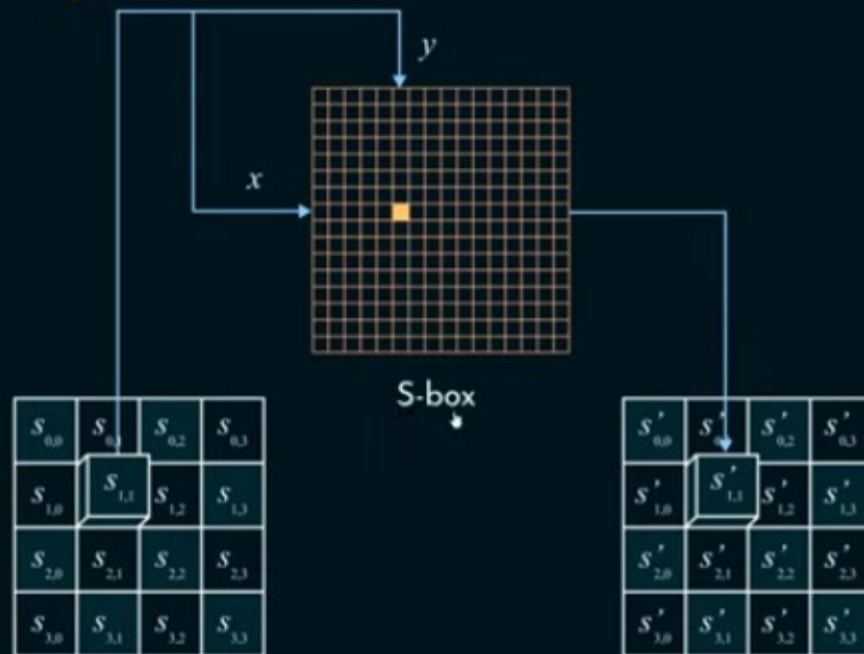
AES Encryption and Decryption



AES Transformation Functions

- ★ Substitute Bytes
- ★ Shift Rows
- ★ Mix Columns
- ★ Add Round Key

Substitute Bytes



AES Sub Bytes

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

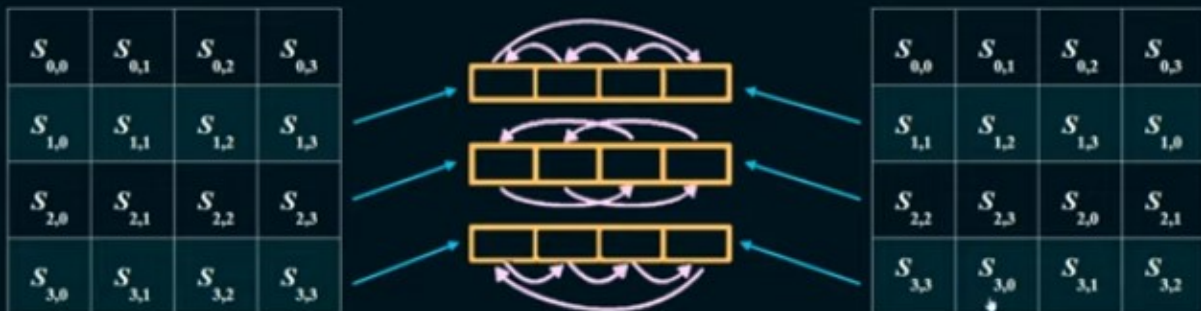
F0 → 8C

EA	04	65	85
83	45	5D	96
5C	33	98	B0
F0	2D	AD	C5

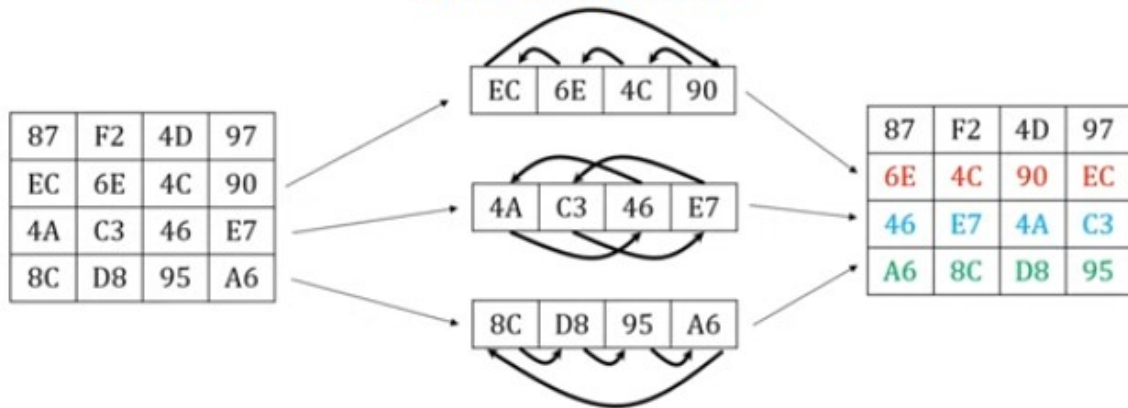


87	F2	4D	97
EC	6E	4C	90
4A	C3	46	
8C			

Shift Rows



AES Shift Rows



Mix Columns



AES Mix Column

2	3	1	1
1	2	3	1
1	1	2	3
3	1	1	2

*

87	F2	4D	97
6E	4C	90	EC
46	E7	4A	C3
A6	8C	D8	95

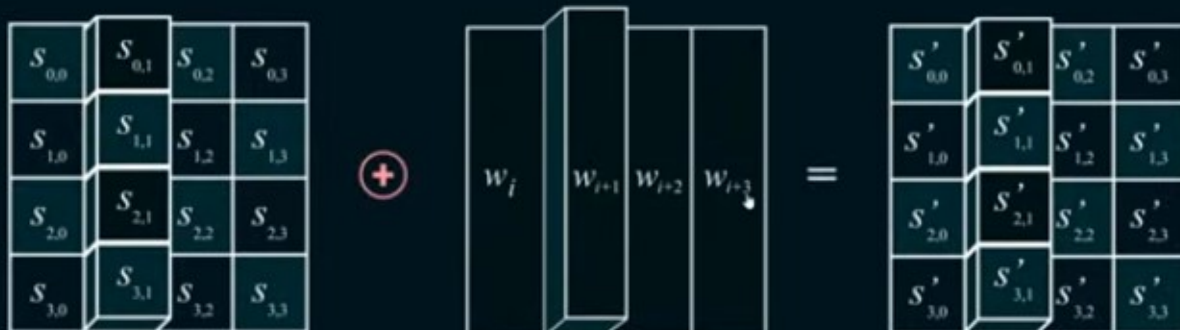
=

?			

$$\{02\} * \{87\} \oplus \{03\} * \{6E\} \oplus \{01\} * \{46\} \oplus \{01\} * \{A6\}$$

$$\begin{aligned}
 & X^7 + X^6 + X^5 + X^4 + X^3 + X^2 + \boxed{X^1} + 1 \\
 & \begin{matrix} \rightarrow 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{matrix} \\
 & 02 = 0000\ 0010 = X \\
 & 87 = 1000\ 0111 = X^7 + X^2 + X + 1 \\
 & \begin{matrix} \boxed{X^7} + X^6 + X^5 + X^4 + X^3 + \boxed{X^2} + \boxed{X^1} + \boxed{1} \\ \begin{matrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{matrix} \end{matrix} \\
 & \{02\} * \{87\} = X * (X^7 + X^2 + X + 1) \\
 & = \boxed{X^8} + X^3 + X^2 + X \\
 & = X^4 + \cancel{X^3} + \cancel{X} + 1 + \cancel{X^3} + X^2 + \cancel{X} \\
 & = X^4 + X^2 + 1 \\
 & = 0001\ 0101 \\
 & \text{Use irreducible Polynomial Theorem, GF}(2^8) \\
 & X^8 = X^4 + X^3 + X + 1 \\
 & \begin{matrix} X^7 + X^6 + X^5 + X^4 + X^3 + X^2 + X^1 + 1 \\ \begin{matrix} 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \end{matrix} \end{matrix}
 \end{aligned}$$

Add Round Key



AES Add Round key

47	40	A3	4C
37	D4	70	9F
94	E4	3A	42
ED	A5	A6	BC

$\oplus$

AC	19	28	57
77	FA	D1	5C
66	DC	29	00
F3	21	41	6A

=

EB			
40			
F2			
1E			

$$47 \oplus AC = ?$$

$$47 = 0100\ 0111$$

$$AC = 1010\ 1100$$

$$\begin{array}{r} 0100\ 0111 \\ 1010\ 1100 \\ \hline 1110\ 1011 \end{array}$$

$$\begin{array}{cc} E & B \end{array} = \{EB\}$$

$$94 \oplus 66 = ?$$

$$94 = 1001\ 0100$$

$$66 = 0110\ 0110$$

$$\begin{array}{r} 1001\ 0100 \\ 0110\ 0110 \\ \hline 1111\ 0010 \end{array}$$

$$\begin{array}{cc} F & 2 \end{array} = \{F2\}$$

$$37 \oplus 77 = ?$$

$$37 = 0011\ 0111$$

$$77 = 0111\ 0111$$

$$\begin{array}{r} 0011\ 0111 \\ 0111\ 0111 \\ \hline 0100\ 0000 \end{array}$$

$$\begin{array}{cc} 4 & 0 \end{array} = \{40\}$$

$$ED \oplus F3 = ?$$

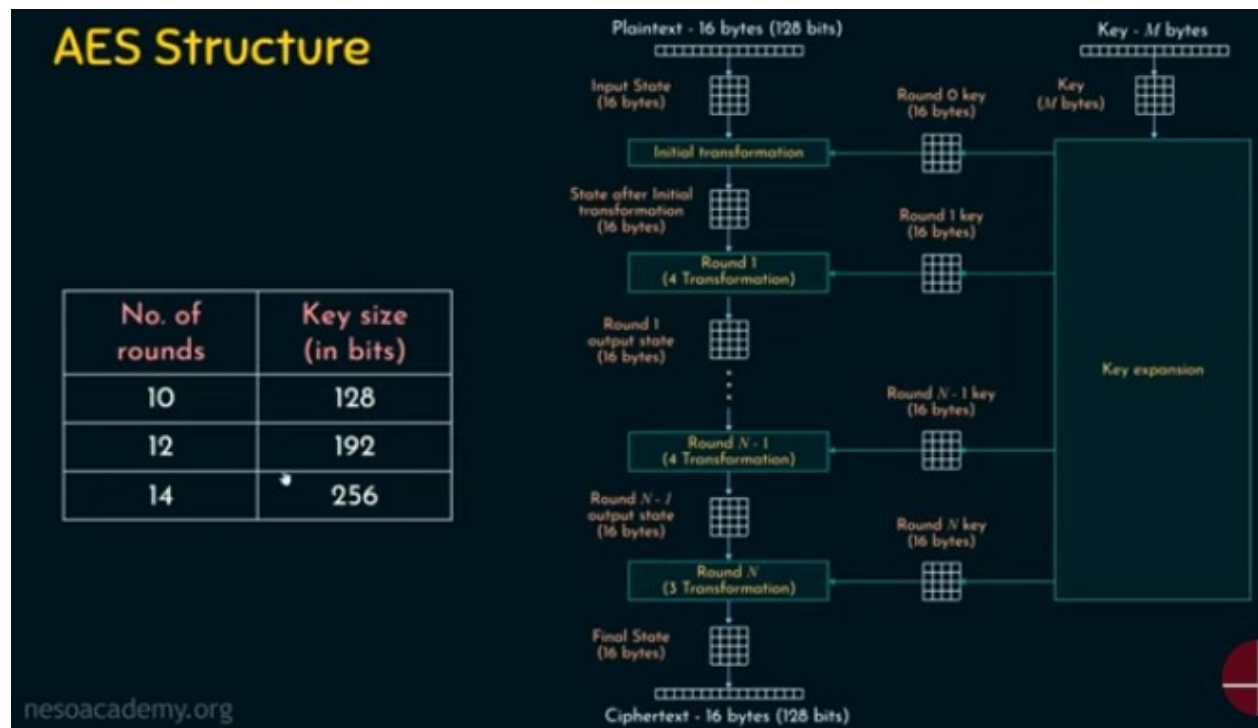
$$ED = 1110\ 1101$$

$$F3 = 1111\ 0011$$

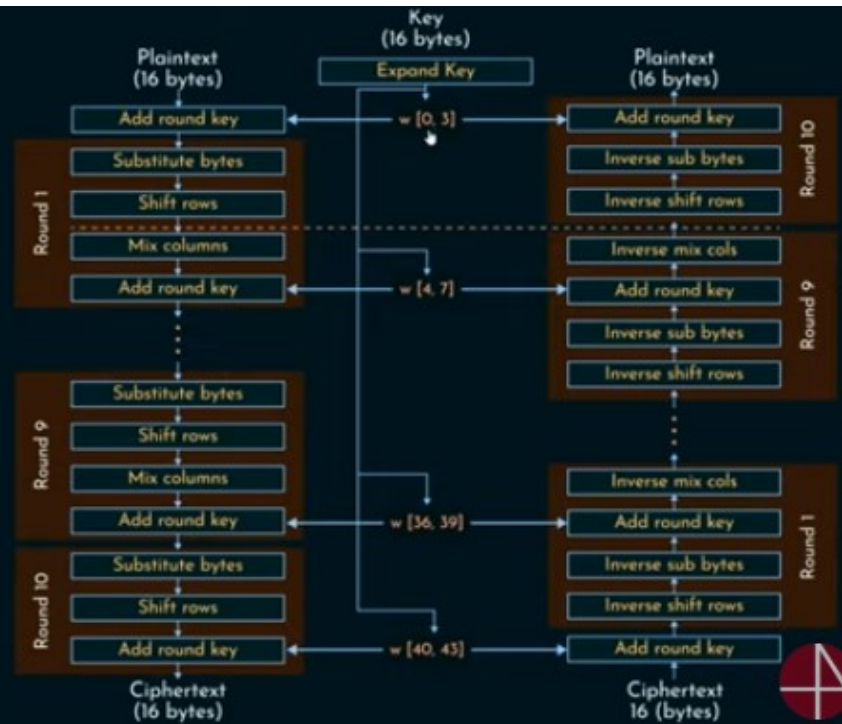
$$\begin{array}{r} 1110\ 1101 \\ 1111\ 0011 \\ \hline 0001\ 1110 \end{array}$$

$$\begin{array}{cc} 1 & E \end{array} = \{1E\}$$

AES Key Expansion



AES Encryption and Decryption

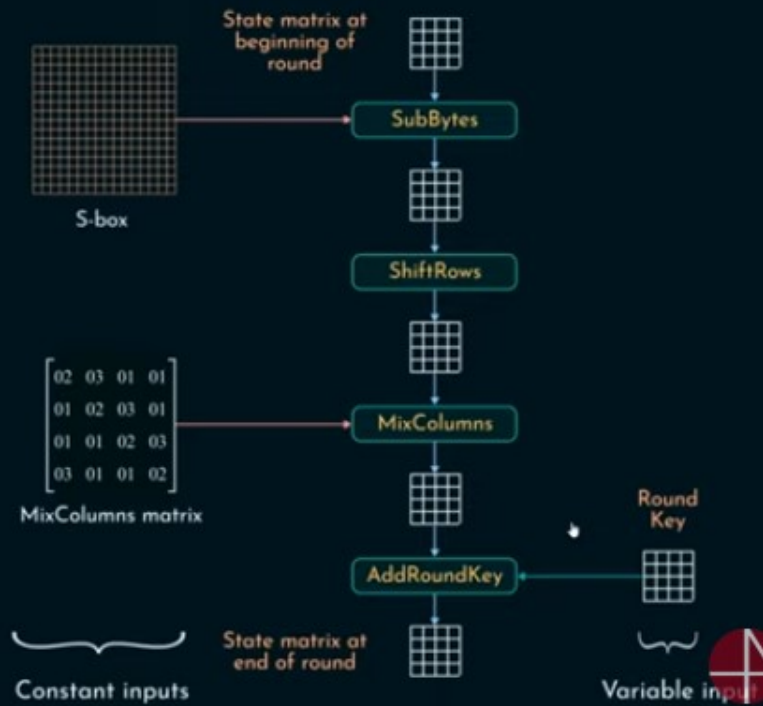


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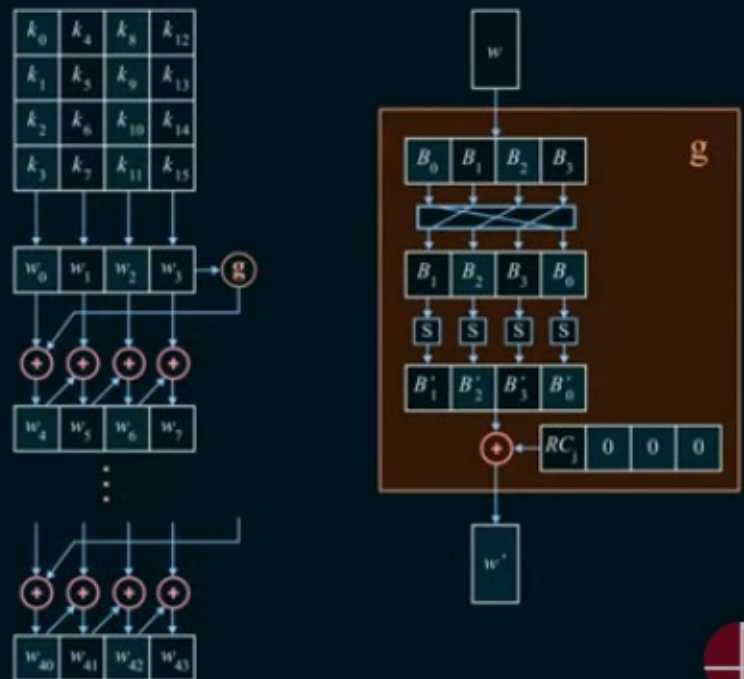
Round Keys in AES 128

Round	Words
Initial Transformation	w_0 w_1 w_2 w_3
Round 1	w_4 w_5 w_6 w_7
Round 2	w_8 w_9 w_{10} w_{11}
...	
Round 10	w_{40} w_{41} w_{42} w_{43}

Inputs for Single AES Round



AES Key Expansion



S-Box in AES

	00	01	02	03	04	05	06	07	08	09	0a	0b	0c	0d	0e	0f
00	63	7c	77	7b	f2	6b	6f	c5	30	01	67	2b	fe	d7	ab	76
10	ca	82	c9	7d	fa	59	47	f0	ad	d4	a2	af	9c	a4	72	c0
20	b7	fd	93	26	36	3f	f7	cc	34	a5	e5	f1	71	d8	31	15
30	04	c7	23	c3	18	96	05	9a	07	12	80	e2	eb	27	b2	75
40	09	83	2c	1a	1b	6e	5a	a0	52	3b	d6	b3	29	e3	2f	84
50	53	d1	00	ed	20	fc	b1	5b	6a	cb	be	39	4a	4c	58	cf
60	d0	ef	aa	fb	43	4d	33	85	45	f9	02	7f	50	3c	9f	a8
70	51	a3	40	8f	92	9d	38	f5	bc	b6	da	21	10	ff	f3	d2
80	cd	0c	13	ec	5f	97	44	17	c4	a7	7e	3d	64	5d	19	73
90	60	81	4f	dc	22	2a	90	88	46	ee	b8	14	de	5e	0b	db
a0	e0	32	3a	0a	49	06	24	5c	c2	d3	ac	62	91	95	e4	79
b0	e7	c8	37	6d	8d	d5	4e	a9	6c	56	f4	ea	65	7a	ae	08
c0	ba	78	25	2e	1c	a6	b4	c6	e8	dd	74	1f	4b	bd	8b	8a
d0	70	3e	b5	66	48	03	f6	0e	61	35	57	b9	86	c1	1d	9e
e0	e1	f8	98	11	69	d9	8e	94	9b	1e	87	e9	ce	55	28	df
f0	8c	a1	89	0d	bf	e6	42	68	41	99	2d	0f	b0	54	bb	16

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Round Constant in AES-128



Follow

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Round	RC	Round	RC
1	(<u>01</u> 00 00 00) ₁₆	6	(<u>20</u> 00 00 00) ₁₆
2	(<u>02</u> 00 00 00) ₁₆	7	(<u>40</u> 00 00 00) ₁₆
3	(<u>04</u> 00 00 00) ₁₆	8	(<u>80</u> 00 00 00) ₁₆
4	(<u>08</u> 00 00 00) ₁₆	9	(<u>1B</u> 00 00 00) ₁₆
5	(<u>10</u> 00 00 00) ₁₆	10	(<u>36</u> 00 00 00) ₁₆

Note: Initial Transformation takes (00 00 00 00)₁₆ as the RC.

AES Security

- ❖ AES was designed after DES.
- ❖ Most of the known attacks on DES were already tested on AES.
- ❖ Brute-Force Attack.
- ❖ Statistical Attacks.
- ❖ Differential and Linear Attacks.

AES Implementation Aspects

- ❖ Simple Algorithms.
- ❖ Resistant against known attacks.
- ❖ Code compactness on many CPUs.
- ❖ Cheap processors and minimum amount of memory.
- ❖ Very efficient.
- ❖ Implementation of AES.

Multiple Encryption and Triple DES

- ★ Drawbacks of DES.
- ★ Potential vulnerability of DES to a brute-force attack.
- ★ AES - Completely a new algorithm.
- ★ Another alternative.
- ★ Use of multiple encryption with DES and multiple keys.

Double DES

